Security Analysis of Distributed Virtual Reality and Physiological Sensor (VR-PHY) Systems

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***Abstract*—Virtual reality (VR) systems that are integrated with physiological sensors such as electroencephalography (EEG), electromyography (EMG), electrocardiogram (ECG) or galvanic skin response (GSR) provide immersive and interactive experiences in industries, including healthcare, education, and entertainment. Despite their increased usage, these systems face significant security challenges, particularly related to sensitive data interceptation, unauthorized access, and exploitation of the system.**

In this paper, we examine these security threats, focusing on data interception, and present mitigation strategies such as end-to-end encryption, biometric authentication, and anomaly detection to secure such interactive VR-Physiological Sensor (VR-PHY) systems. Through a real-world implementation, we demonstrate that attackers can intercept comprehensive user data streams, capturing precise 3D spatial coordinates, physiological states including heart rate and heart rate variability, and behavioral markers from unencrypted Lab Streaming Layer communications. Our experimental validation reveals that VR headsets serve as primary attack vectors, with vulnerabilities that enable real-time data interception and potential environmental manipulation. We present and evaluate mitigation strategies, including end-to-end encryption, biometric authentication, and anomaly detection, to secure interactive VR-PHY systems.

***Index Terms*—Virtual Reality(VR), Electroencephalogram (EEG), Electromyography (EMG), Electrocardiogram (ECG), Galvanic Skin Response (GSR), Security**

I. Introduction

The integration of physiological sensors with VR systems has created a revolution in human-computer interaction, offering highly personalized and immersive experiences to users. These types of system have evolutionary applications in healthcare, rehabilitation, gaming, and education [4]. As VR systems increasingly integrate physiological sensing, the collected data extend beyond conventional interaction metrics and enter the domain of biometric and health-related information. Signals such as heart rate variability, electrodermal activity, EEG recordings, and motion trajectories can reveal stress levels, cognitive workload, emotional states, and behavioral patterns. Such multimodal data are often classified as sensitive personal data under regulatory frameworks such as HIPAA and GDPR when deployed in healthcare or clinical contexts.

Prior research demonstrated that VR motion data can be revese-engineered to infer keystrokes, speech content, and even user identity through side-channel attacks [1] [10] [12]. Similarly, work in eHealth monitoring systems has shown that intercepted physiological data can expose daily routines, medical conditions, and location patterns [2], alongside replaying normal physiological data to prevent a system from raising an alarm [9]. These findings demonstrate that the risks are not hypothetical. As VR-PHY systems move toward real-world deployment in rehabilitation, pilot training, and adaptive therapy platforms, securing their data transmission becomes a timely and necessary requirement.

At the architectural level, VR-PHY systems consist of three primary components: a VR headset, physiological sensing devices (e.g., EEG, EMG, ECG, GRS), and synchronization middleware such as Lab Streaming Layer (LSL), which aligns multimodal data streams in real time [3]. Because the VR headset acts as the central hub for data aggregation and communication, it becomes a critical attack surface for data interception, malware injection, and motion-based inference attacks [12].

This paper provides a system-level security analysis of distributed VR-PHY architectures, demonstrating how default deployments expose synchronized biometric and spatial data to interception and misuse. Figure 1 illustrates the high-level structure of the VR-PHY system, integrating data acquisition, synchronization, application processing, and visualization modules. Physiological sensors capture neural and bodily signals, while the VR headsets record spatial position and orientation data. These heterogeneous data streams are synchronized through LSL and consumed by Unity-based applications for visualization and adaptive interaction [3].

Despite their transformative potential, VR-PHY systems introduce systemic security risks. Unsecured LSL streams, wireless sensor transmissions, and third-party SDK integrations may allow data interception, unauthorized access, or

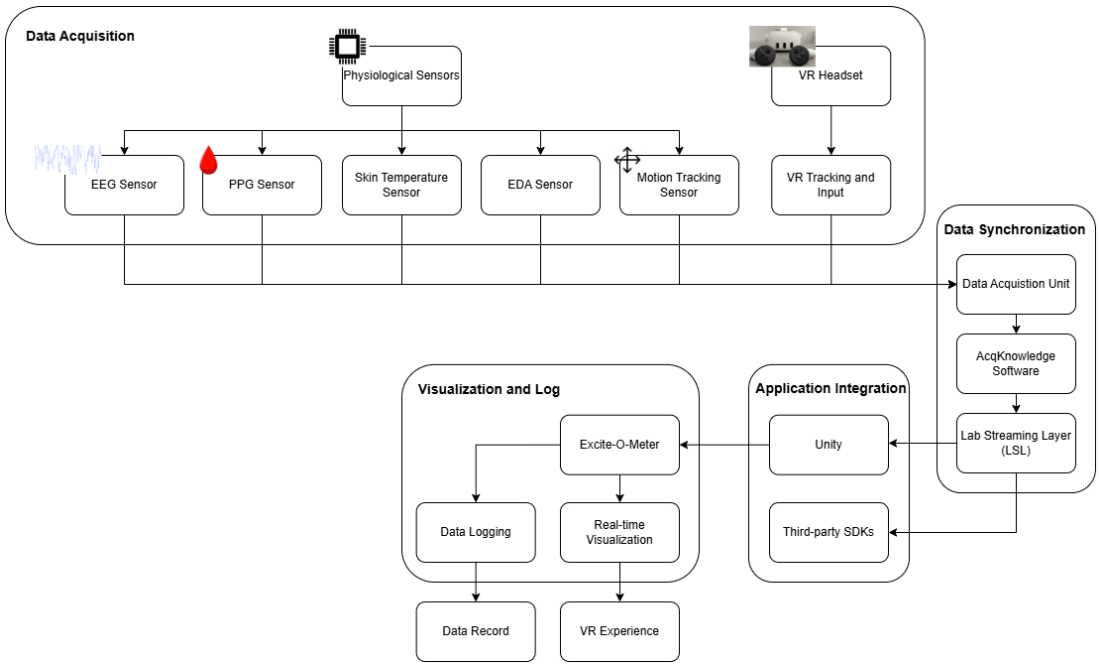

Fig. 1: High-Level System Architecture in VR-PHY Systems

manipulation of biometric signals. If physiological data are stored or transmitted without encryption or access control, sensitive health-related information may be exposed. Furthermore, reliance on external APIs and SDKs increases the attack surface through potential misconfigurations, outdated libraries, or weak authentication mechanisms.

Figure 2 demonstrates the system-level connection of physiological sensing, emotion analysis, and VR technologies to create an emotionally adaptive user experience. Physiological sensors such as EEG headsets, BIOPAC Research Rings, or other proprietary devices capture real-time biometric signals from users. Now these signals are processed through a dedicated sensor software that helps to translate raw physiological input to usually digital data. A Python script acts as an interface to the software to clean, pre-process, and extract features from the incoming data, allowing structured input downstream for the emotion analysis modules.

The emotion analysis component engages computational models to infer emotional states such as stressed, calm, or excitement based on input from physiological sensory devices. These states are also displayed through an Excite-o-Meter [7]. This component helps to display the sensory data in the VR environment and is also used to dynamically alter the VR environment based on the user's physiological condition. In addition, an AI assistant interacts with the user on their inferred emotional state, offering adaptive support and personalized suggestions or cognitive interventions to enhance the user experience.

II. Literature Review

In this section, we provide a review of the literature on security issues in the two main parts of the VR-PHY system:

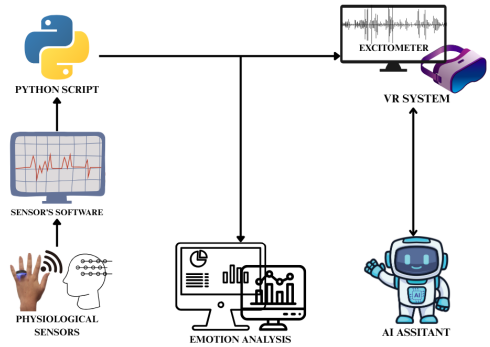

Fig. 2: A real-world implementation of VR-PHY system for Emotion Adaptive VR Experience

VR headsets and physiological sensors.

A. VR Headsets

VR headsets act as the central hub of VR-PHY systems, interacting seamlessly with EEG sensors, physiological monitors, and external networks. Despite that, their critical role opens them up to various security vulnerabilities, especially in wireless communication.

Data interception is one such significant concern, as VR headsets transmit sensitive information such as neural activity, motion data, and physiological metrics via wireless channels such as WiFi or Bluetooth. If there are no proper encryption protocols, attackers can easily intercept and exploit these data streams. For example, the VR-Spy attack showed how WiFi Channel State Information (CSI) could be used to infer

user actions, such as virtual keystrokes, by analyzing motion patterns during transmission. This vulnerability can lead to identity theft, unauthorized access, or manipulation of VR experiences [1].

Another critical vulnerability is the exploitation of motion data. Embedded motion sensors, such as accelerometers and gyroscopes, can be exploited for side-channel attacks. The Face-Mic attack showed how the subtle facial dynamics captured by zero-permission sensors in VR headsets could also infer private user information such as speech content, identity, and even gender. This capability not only breaches user privacy, but also creates the way for advanced profiling and surveillance threats [10].

Furthermore, malware injection and weak authentication mechanisms exacerbate VR headset vulnerabilities due to its connectivity to the Internet. VR headsets are also vulnerable to malicious software that can manipulate VR environments, exfiltrate data, or manipulate user perceptions. Devices shared in public or multi-user settings often lack robust authentication, which often leads users to unauthorized access.

We can notice how motion sensor data collected by VR systems can be correlated with user behaviors in the physical world to deanonymize individuals [8]. With the help of a machine learning framework, researchers demonstrated how the movements of virtual avatars could be matched with real-world motion data, exposing users to identification risks. Such types of de-anonymization techniques are particularly concerning for users in shared virtual spaces because attackers can stealthily gather auxiliary motion data to profile and identify them. They also emphasize the importance of anonymizing motion data and implementing frameworks to securely handle and process these data sets to protect user identities.

B. Physiological Sensor Device

Physiological sensors, such as BIOPAC Research Rings, face several critical vectors of attack that compromise system reliability and user safety. For instance, malicious actors can manipulate recorded health metrics via false data injection, deceiving monitoring systems and potentially triggering incorrect clinical responses [5]. Wireless sensor networks are particularly susceptible to sinkhole attacks, a scenario in which a malicious node masquerades as the optimal routing path to intercept, drop, or alter packets before they reach the sink [11].

Furthermore, unauthorized access to continuous physiological data allows attackers to deduce a user's daily routines, physical location, and periods of vulnerability [2]. Such longitudinal tracking facilitates stalking, harassment, or targeted physical threats. Finally, attackers may intercept communication channels to extract sensitive metrics. By employing brute-force decryption or password-guessing techniques, adversaries can gain unauthorized access to encrypted streams, leading to identity theft or medical fraud [6].

III. Experimental Validation of VR-PHY System Attack Vectors

A. Attacker Model

To systematically evaluate the security of the VR-PHY architecture, we define an attacker model that reflects realistic deployment conditions in laboratory and prototype environments.

We assume an adversary with network-adjacent access, meaning the attacker is connected to the same local wireless network as the VR headset, physiological sensors, and processing workstation. The attacker does not require physical access to devices and does not break cryptographic primitives. Instead, the attacker leverages unencrypted communication channels and weak authentication mechanisms within the default VR-PHY pipeline.

The adversary is capable of:

- Passive packet interception and traffic analysis of LSL streams.
- Replay or injection of unverified packets.
- Exploitation of insecure third-party SDK integrations within Unity.
- Reconstruction of synchronized motion and physiological logs.

The attacker's objectives are to:

- Capture synchronized biometric and spatial data streams (e.g., HR, HRV, EEG/EDA signals).
- Infer behavioral, emotional, or identity-related information from multimodal data.
- Enable potential downstream manipulation in systems that adapt VR environments based on physiological feedback.

Figure 3 shows the attacker's observation and injection points within the distributed VR-PHY architecture. The adversary is positioned at the network and middleware layers, where communication between physiological sensors, LSL synchronization, Unity integration, and visualization components occurs. As shown in Figure 3, attack vectors include network-level packet sniffing of LSL streams, malicious SDK/API injection within the integration layer, and plaintext session log extraction at the visualization and storage stage. Yellow dashed paths indicate the categories of exposed data, including synchronized biometric metrics and event markers.

B. Vulnerability Mapping

In Figure 4, the flow chart shows an overview of the components and possible vulnerabilities inherent in the VR-PHY systems using our real-world lab implementation of the system. Starting from the VR headset and physiological sensors, the data flows through custom pre-processing logic, LSL outlets, Unity inlets, and processing units before reaching client devices and external APIs. Here, each connection in the system is labeled with its corresponding vulnerabilities, such as data interception, unencrypted communication, packet injection, and unauthorized access. These vulnerabilities show the critical need for robust encryption protocols, secure API

TABLE I: Detailed Explanation of sessionData.json Fields

Field Name	Explanation	Example Value
Timestamp	Time in seconds since the start of the session. Helps align biometric and positional data.	1.281
Headset Position (posX/Y/Z)	3D position of the headset in Unity coordinates. Used to track user movement.	-0.01, 1.52, 0.03
Velocity (velPosX/Y/Z)	Velocity of the headset on each axis. Zero may indicate no motion or a sensor issue.	0.0, 0.0, 0.0
Acceleration (accPosX/Y/Z)	Acceleration of the headset on each axis. Useful for detecting sudden movements.	0.0, 0.0, 0.0
Quaternions (q0, qi, qj, qk)	Four-dimensional complex numbers representing headset orientation and gaze direction.	1.0, 0.0, 0.0, 0.0
Heart Rate (HR)	Number of heart contractions per minute; reflects physiological arousal or stress.	69
RR Interval (RR)	The time between successive R-waves of the ECG (heartbeats) in milliseconds.	872.5
Root Mean Square of Successive Differences (RMSSD)	A time-domain HRV metric reflecting short-term parasympathetic activity.	45.6
Standard Deviation of NN Intervals (SDNN)	The standard deviation of all RR intervals. Reflects overall heart rate variability.	30.2
Extraocular Muscle (EOM/EMG)	Measure of eye or facial muscle electrical activity. NaN may indicate sensor failure.	NaN
Manual Event Markers	List of manually set session markers (e.g., experiment start or task switch).	["ExperimentStarted"]
Screenshot Timestamps	Recorded times when visual captures were taken for post-session data correlation.	[1.281]

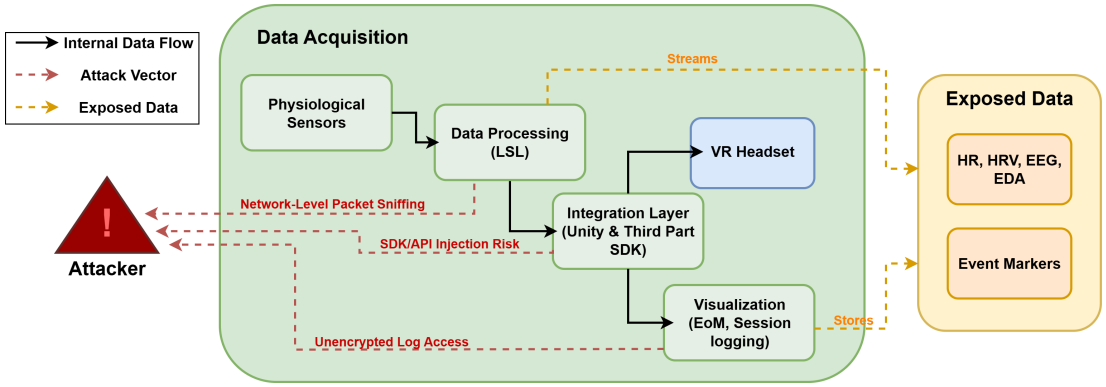

Fig. 3: Attacker Observations and Injection Points in VR-PHY

implementations, and rigorous authentication mechanisms at every stage of data transfer and processing. The flowchart representation emphasizes the complexity of securing these systems, in which each component and communication channel has unique challenges.

Third-party SDKs collect physiological data such as eye movements, hand movements, and EEG signals. These SDKs provide APIs that Unity uses to process incoming data. Once processed in Unity, the data are transmitted via the LSL to the Excite-O-Meter, which visualizes the information in real time and displays biometric and interaction data, enabling the monitoring of user states and VR activity.

Data transmission between Unity, SDKs, and Excite-O-Meter is vulnerable to several security threats. These include man-in-the-middle attacks due to lack of encryption, unauthorized access from weak authentication, and malware/code injection risks from unverified third-party SDKs. Attackers may exploit external scripts or logs in Unity or use compromised SDKs to introduce harmful code. Additionally, hardware and sensor vulnerabilities—such as sensor spoofing or device hijacking—can allow attackers to manipulate or fake biometric data, leading to false outputs in the Excite-O-Meter.

The LSL plays a central role in the processing, synchronizing, and streaming of physiological sensor data in real time. Sensors like EEG caps and BIOPAC devices collect data (e.g., EEG, EDA, motion) and transmit them via USB, Bluetooth, or Wi-Fi. LSL ensures time alignment across these signals and sends the synchronized data to Unity, where it is visualized by the Excite-O-Meter. APIs and third-party SDKs enable communication with external devices, such as VR headsets, allowing real-time biofeedback and adaptive user interactions.

Several vulnerabilities exist in the connection between LSL data processing and external devices. In data processing, unencrypted data transmission can occur if EEG caps and

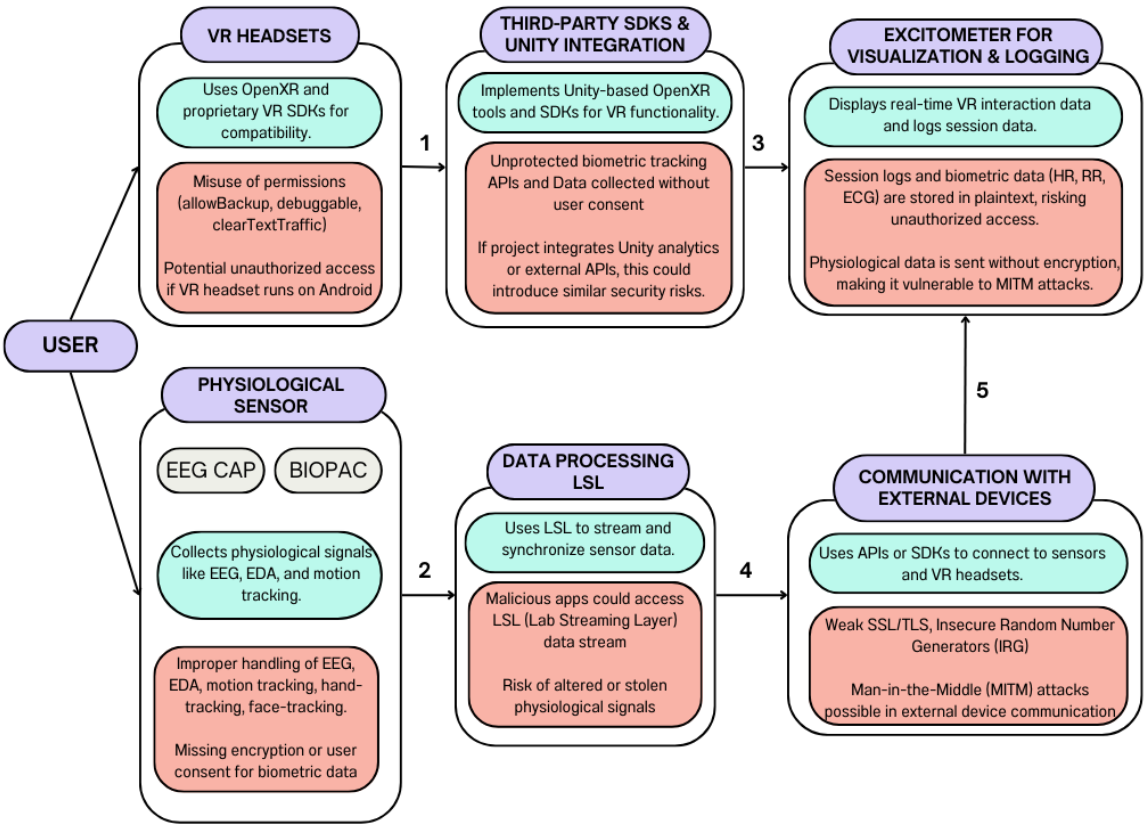

Fig. 4: Security Vulnerabilities and Risk Assessment in VR-PHY Systems

Biopac sensors transmit data over Bluetooth or Wi-Fi, making it susceptible to interception by malicious actors. Device hijacking is also a risk if Bluetooth or USB drivers have weak security, allowing attackers to take control of biometric sensors or modify the output of the data. Unauthorized access can occur if the sensors lack proper authentication, allowing unauthorized users to access sensitive physiological data.

In data streaming, vulnerabilities can lead to data interception and eavesdropping. If encryption is not enforced, attackers on the same network can read or alter real-time biometric data. Data transmission from the LSL to the Excite-O-Meter and Unity also presents risks, including data logging, man-in-the-middle attacks, and insecure API calls. If the Excite-O-Meter stores session logs and biometric data in plaintext, it becomes vulnerable to unauthorized access. For man-in-the-middle attacks, if physiological data is transmitted without encryption, attackers could alter the data before it reaches Unity. Insecure API calls also pose a threat, as Unity receives biometric data through APIs, giving attackers the opportunity to exploit vulnerabilities to steal or manipulate data.

Communication with external devices can have vulnerabilities if APIs or SDKs use weak or outdated encryption methods, making data exchanged between devices susceptible to compromise. In addition, unsecured external device communication can make it vulnerable to man-in-the-middle attacks, allowing attackers to intercept biometric data while in transit.

IV. Experimental Setup

In this section, we discuss the setup of the VR-PHY system in order to evaluate the security and integrity of data transmission.

A. Hardware Layer

The setup consists of a Meta Quest 2 VR headset and a Senstream Research Ring. The Research Ring is used to collect the PPG, EDA, ECG, skin temperature, and motion of the user, as shown in Figure 5. These devices work in tandem to transfer user data over to the software layer.

B. Software and Middleware Layer

The Research Ring data is transmitted over to a custom Unity program called the Excite-o-Meter, which calculates and displays real-time physiological metrics, such as heart rate variability (HRV) or arousal levels. In order to do so, the Lab Streaming Layer synchronizes the Senstream Research Ring and the Meta Quest 2 and converts the data into a readable format for the Excite-o-Meter.

The Lab Streaming Layer was built mainly for convenience, not security, meaning it does not include by default any encryption or authentication. This allows for us to test what kind of information we can see from the outside.

datetime	sessionTime	posX	posY	posZ	q0	qi	qj	qk	pitch	yaw	roll
2.41E+11	0.12	0	1	-4.616	1	0	0	0	0	0	0
2.41E+11	0.24	0	1	-4.616	1	0	0	0	0	0	0
2.41E+11	0.36	0	1	-4.616	1	0	0	0	0	0	0
2.41E+11	0.48	0	1	-4.616	1	0	0	0	0	0	0
2.41E+11	0.6	0	1	-4.616	1	0	0	0	0	0	0
2.41E+11	0.72	0	1	-4.616	1	0	0	0	0	0	0
2.41E+11	0.84	0	1	-4.616	1	0	0	0	0	0	0
2.41E+11	0.96	0	1	-4.616	1	0	0	0	0	0	0
2.41E+11	1.08	0	1	-4.616	1	0	0	0	0	0	0

TABLE II: A snippet of motion data found within the VR headset.

datetime	sessionTime	velPosX	velPosY	velPosZ	accPosX	accPosY	accPosZ	velQ0	velQi	velQj
2.408E+11	0.12	0	0	0	0	0	0	1	0	0
2.408E+11	0.24	0	0	0	0	0	0	1	0	0
2.408E+11	0.36	0	0	0	0	0	0	1	0	0

velQk	accQ0	accQi	accQj	accQk	velPitch	velYaw	velRoll	accPitch	accYaw	accRoll
0	1	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0

TABLE III: A snippet of movHeadsetFeatures.csv: Extracted motion features (Split due to width constraints)

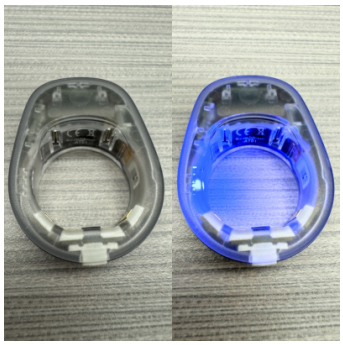

Fig. 5: The Senstream Research Ring

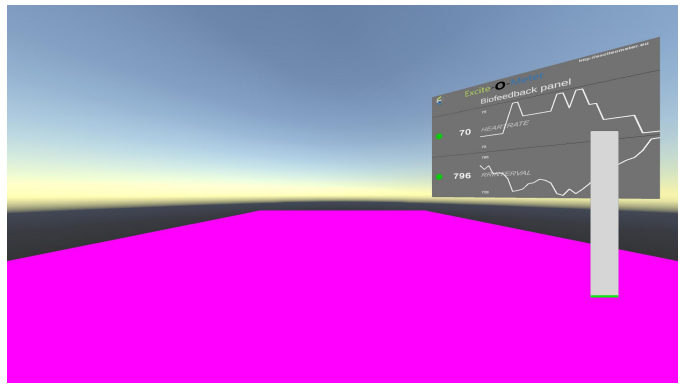

Fig. 6: Runtime screenshot: Live VR session snapshot

datetime	sessionTime	type	message	details
2.41E+11	0	ManualMarkers	ExperimentStarted	CUSTOM_MARKER
2.41E+11	0	Screenshots	screen_1920x1080_	NONE

TABLE IV: eventsAndMarkers.csv: Event and marker log

V. Experimental Results and Risk Analysis

In this section, we summarize the results of our experiment. Logs have been generated to analyze various features of the VR-PHY system. The log of the main session captures timestamped records of:

- **Headset position, velocity, and acceleration,** shown in Table II.
- **Motion features**, shown in Table III.
- **Physiological data:** Heart rate (HR), RR interval (RRi), root mean square of successive differences (RMSSD), standard deviation of normal to normal intervals (SDNN), extraocular muscles (EOM), shown in Table I.
- **Manual markers** (e.g., "ExperimentStarted"), shown in Table IV.

The storage mechanisms used for these physiological logs—such as inadequate encryption or weak access controls—create a vulnerability that an attacker can exploit to gain unauthorized access and exfiltrate these sensitive data. Such exploits may also lead to unintended spying on live VR sessions, as shown in Figure 6.

A. Interception Analysis

Our experimental validation demonstrated the feasibility of intercepting sensitive data streams within the VR-PHY system. The analysis of these intercepted logs shows the system's vulnerabilities and the concrete risks to user privacy and security. The comprehensive nature of the captured data is detailed in Table I.

The interception captured multiple synchronized data streams as shown in Tables II and III, the headset motion log provided continuous tracking of the user's precise 3D position (e.g., -0.01, 1.52, 0.03), velocity, acceleration, and orientation quaternions. These spatial data were complemented by the extraction of high-level motion characteristics, as seen in Figure 5, which could be used to infer specific use actions and behaviors.

During the interception, critical physiological and event

	Use Case	Issue	Cause of Issue	Probability	Consequence
1	VR Headset Connectivity	Data Transmission Delay	Network Latency	2	3
2	VR Wireless Streaming	Data Loss Due to Connection Issues	Poor Wi-Fi Signal / Weak API Integration	3	4
3	VR Data Communication	Man-in-the-Middle (MITM) Attack	Lack of Encryption in Wireless Transmission	3	4
4	VR Runtime / Platform Layer	API Exploits or Misconfigurations	Improper Access Controls in OpenXR/SteamVR	2	4
5	Physiological Sensor Security	Sensor Spoofing	Lack of Sensor Authentication	1	4
6	Biometric Data Logging	Plaintext Storage of Biometric Data	Lack of Encryption for Stored Data	2	5
7	Data Transmission via LSL	Unauthorized Access to Physiological Data	Weak Authentication in LSL Data Streams	2	2
8	Third-Party SDKs in Unity	Insecure or Outdated SDKs	Lack of Security Patches for SDKs	2	2
9	External Device Communication	Weak SSL/TLS Implementation	Outdated or Misconfigured SSL/TLS Protocols	2	3
10	VR Application Plugins	Untrusted Plugin Code	Lack of Code Validation	2	3
11	LSL Data Transmission	Data Interception and Eavesdropping	Lack of Encrypted Data Streams	2	3
12	Excite-O-Meter Logging	Lack of Integrity Verification in Logs	No Cryptographic Hashing or Access Control	2	3
13	Communication with External Devices	Man-in-the-Middle (MITM) Attacks	Unsecured Device-to-Device Communication	1	3
14	Wireless Sensor Transmission	Unencrypted Physiological Data	Lack of Secure Data Transmission Protocols	2	3
15	VR Headset API Security	Lack of User Consent for Biometric Data	No Consent Mechanism in SDK	2	4
16	VR Application Debug Mode	Debug Builds in Production	Enabled Debugging Tools in Release Version	3	2

TABLE V: VR-PHY Security Risk Summary. Each risk is scored on a scale of 1 to 5, 1 having the least amount of risk/chance to happen and 5 having the most amount of risk/chance to happen.

data were captured. The system recorded sensitive biometric information, including heart rate (69 BPM), RR interval (872.5 ms), and heart rate variability metrics (RMSSD: 45.6, SDNN: 30.2). Furthermore, as evidenced in Table IV, the attack successfully captured temporal markers such as the event "ExperimentalStarted" at the 1.281-second timestamp, creating a precise timeline of user interaction.

B. Risk Assessment and Discussion

In this section, we present a detailed analysis of the risk assessment performed for the integrated VR-PHY system, as shown in Table V.

The risk assessment process began by identifying key vulnerabilities in the VR-PHY system through collaboration between developers and the security team. The risks were related to specific use cases, functional components of the system, such as wireless data transmission from physiological sensors. Each risk was described in terms of the issue, its cause and its potential impact. Examples include data intercept due to lack of encryption and sensor spoofing from weak security protocols, as seen in Table V. Risks were rated according to their likelihood and impact to help prioritize mitigation strategies.

In the probability evaluation, the likelihood that each risk occurs is rated on a scale from 1 to 5, where 1 represents a rare occurrence and 5 signifies a frequent or almost certain occurrence. The following ratings were applied based on the system analysis and data, where available:

1. Probability: Chance of risk:

- (1) Rare - slim possibility of happening.
- (2) Unlikely – Could occur but remains remote.
- (3) Possible - might occur under certain conditions.
- (4) Expected to occur in most deployments.
- (5) Almost certain; expected regularly or observed frequently.

The consequence rating evaluates the potential impact of each identified risk on the performance, security, and user experience of the VR-PHY system. The following scale was used to assess the consequences.

2. Consequence: Impact on the project:

- (1) Insignificant – Not worth worrying about; negligible impact.
- (2) Minor – May cause delays or minor added costs.
- (3) Moderate - likely to introduce delays or increase resource requirements.
- (4) Major - can lead to partial or complete system failure.
- (5) Severe, definitive failure or major breach; possible system compromise.

The probability and consequence scores reported in Table V were derived through a qualitative security assessment of the VR-PHY architecture conducted by the system developers and the security analysis team. Probability values were estimated based on observed interception feasibility in the experimental environment, the exposure of communication channels identified in Section III, and the likelihood of exploitation in typical

VR-PHY deployments. Consequence values were assigned according to the potential impact of each vulnerability on system integrity, user privacy, and operational reliability, particularly considering the sensitivity of physiological and behavioral data processed by the platform. This expert-driven evaluation allowed the development team to prioritize security risks based on both architectural exposure and potential real-world impact.

The comprehensive risk assessment reveals that VR-PHY systems face their most critical threats in the data pipeline architecture, where authentication weaknesses and unencrypted communications create systemic vulnerabilities. The high concentration of risks with consequence ratings of 4 or 5, particularly in biometric data storage (#6), sensor authentication (#5), and wireless transmission (#3, #14), demonstrates that the core data handling mechanisms of the system represent the primary attack surface. This pattern indicates that security investments should prioritize cryptographic protection of physiological data streams and implement robust device authentication protocols before addressing vulnerabilities of lower impact. The assessment provides a clear roadmap for security hardening, emphasizing that protecting the integrity of biometric data flows must be the foundational priority for any VR-PHY deployment in sensitive environments.

C. Discussion and Limitations

While this study demonstrates the feasibility of intercepting synchronized biometric and spatial data in a distributed VR-PHY architecture, several limitations must be acknowledged. The experimental validation was conducted within a controlled laboratory prototype environment. Although the attacker model reflects realistic network-adjacent conditions, the results may differ in production deployments that implement stronger encryption, hardened network segmentation, or enterprise-grade authentication controls. This work focuses primarily on architectural vulnerability analysis and interception feasibility rather than quantitative performance benchmarking. We did not measure packet loss rates, interception success probabilities under varying network conditions, or computational overhead introduced by potential mitigation strategies.

The attacker model assumes a network-adjacent adversary without compromising physical devices. More advanced threat actors, including those with firmware-level access or hardware tampering capabilities, were not evaluated in this study. Despite these limitations, the findings highlight structural security weaknesses inherent in default VR-PHY integrations and provide a foundation for future quantitative and defensive research.

VI. Conclusion and Future Directions

The integration of distributed VR-PHY systems represents a transformative advancement in interaction simulations and human-centered computing, yet it introduces unprecedented security challenges that demand immediate attention. Our analysis demonstrates that these systems create a complex attack surface where physiological data streams, spatial tracking information, and behavioral metrics converge, creating multiple vectors for exploitation. The most critical vulnerabilities arise from the fundamental architecture of these systems, leaving sensitive biometric and motion data exposed to interception and manipulation.

The security implications extend beyond conventional data breaches to include psychological privacy and physical safety. The ability to correlate physiological responses with virtual experiences enables sophisticated behavioral profiling that could lead to unauthorized psychological assessment and targeted manipulation. Furthermore, manipulation of biometric feedback or spatial coordinates could induce physical discomfort, simulator sickness, or even hazardous situations in safety-critical applications.

Looking ahead, securing VR-PHY systems requires a multi-layered approach that addresses both technical and human factors. Future work should incorporate quantitative evaluation to assess scalability and resilience under diverse deployment scenarios, such as implementing data transmission protection in the VR-PHY system via encryption or authentication.

References

[1] Abdullah Al Arafat, Z. Guo, and A. Awad. VR-Spy: A Side-Channel Attack on Virtual Key-Logging in VR Headsets. In *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, March 2021.

[2] F. Kargl, E. Lawrence, M. Fischer, and Y. Y. Lim. Security, Privacy and Legal Issues in Pervasive eHealth Monitoring Systems. In *IEEE Conference on Intelligent Sensors, Sensor Networks and Information Processing (ISSNIP)*, July 2008. Accessed: Jan. 10, 2023.

[3] C. Kothe et al. The Lab Streaming Layer for Synchronized Multimodal Recording. *bioRxiv*, February 2024.

[4] H. Li, H. Shin, L. Sentis, K.-C. Siu, J. del R. Millán, and N. Lu. Combining VR with electroencephalography as a frontier of brain-computer interfaces. *Device*, 2(6):100425, June 2024.

[5] T. Ni, Y. Du, Q. Zhao, and C. Wang. Non-intrusive and Unconstrained Keystroke Inference in VR Platforms via Infrared Side Channel, 2024. Accessed: Oct. 10, 2025.

[6] A. Omolara, A. Jantan, O. Abiodun, H. Arshad, K. Dada, and E. Emmanuel HoneyDetails. A prototype for ensuring patient's information privacy and thwarting electronic health record threats based on decoys. *Health Informatics Journal*, 26(3):2083–2104, 2020.

[7] L. Quintero, J. E. Munoz, Jeroen de Mooij, and M. Gaebler. Excite-O-Meter: Software Framework to Integrate Heart Activity in Virtual Reality. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, October 2021.

[8] M. A. Sabra. *Addressing Privacy Challenges in Modern Audio-Video Communication Systems and Applications*. Ph.d. dissertation, The University of Texas at San Antonio, May 2023.

[9] O. Salem, C. Fehri, W. Itani, S. Hamrioui, J. Boubeta-Puig, and R. Boutaba. Man-in-the-middle attack mitigation in internet of medical things. *IEEE Transactions on Industrial Informatics*, 17(10):7051–7061, 2021.

[10] C. Shi et al. Face-Mic: Inferring Live Speech and Speaker Identity via Subtle Facial Dynamics Captured by AR/VR Motion Sensors. In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2022.

[11] F. Sun, Z. Zhao, Z. Fang, L. Du, Z. Xu, and D. Chen. A Review of Attacks and Security Protocols for Wireless Sensor Networks. *Journal of Networks*, 9(5), May 2014.

[12] Z. Yang, C. Y. Li, A. Bhalla, B. Y. Zhao, and H. Zheng. Inception Attacks: Immersive Hijacking in Virtual Reality Systems, March 2024.